
Session 2 — Statistics & Inference

What today is about

Last session we studied *single* uncertain quantities. Today we turn to *data*: many observations at once, and what we can learn from them.

- Summarise data with a few numbers: *mean, variance, covariance*.
- Understand why averaging works: the *Law of Large Numbers* and the *Central Limit Theorem*.
- Infer from a sample to a population: *standard errors, confidence intervals, hypothesis tests*.
- Build our first real model: *linear regression*.

The thread running through everything

Population vs. sample

There is a fixed but unknown truth about the whole *population* (for example, the true average reaction time of all humans). We only ever see a limited *sample*. Statistics is the disciplined art of guessing the population from the sample — and honestly reporting how uncertain that guess is.

Keep asking, on every slide: is this a statement about the population, or about our particular sample?

The centre of a distribution: expectation

The first thing we want to know about data is where it is centred.

Definition

The **expectation** (mean) is the probability-weighted average value:

$$\mathbb{E}[X] = \sum_x x p(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x f(x) dx \quad (\text{continuous}).$$

Picture the distribution as a pile of mass on the number line: the expectation is its *balance point*, where a finger would keep it level. It need not be a value the variable can actually take — a fair die has mean 3.5.

Expectation adds up nicely

The property that makes expectation so useful is *linearity* — and it holds always, even when variables influence each other.

Useful fact

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

Example

A fair die has $\mathbb{E}[X] = \frac{1}{6}(1 + 2 + \cdots + 6) = 3.5$. A game paying $Y = 2X + 1$ points has $\mathbb{E}[Y] = 2(3.5) + 1 = 8$ — no need to re-average the six payouts.

How spread out is the data? Variance

The second thing we want is how much the data spreads around its centre.

Definition

The **variance** is the average squared distance from the mean, and the **standard deviation** is its square root (back in the original units):

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \text{SD}(X) = \sqrt{\text{Var}(X)}.$$

We square the deviations so positive and negative ones do not cancel, and to penalise large departures more. The square root then brings us back to sensible units — milliseconds, not milliseconds-squared.

Three variance rules worth memorising

Useful fact

Computational formula: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ — usually the fastest by hand.

Shift and scale: $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (adding a constant does not change spread).

Sum: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

That last term — the covariance — is exactly the correction for when X and Y move together. It vanishes only when they are uncorrelated.

Example

Find the variance of a fair die.

Using the computational formula, we need the mean of the square:

$$\mathbb{E}[X^2] = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \frac{91}{6} \approx 15.17.$$

Then

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 15.17 - 3.5^2 = 2.92, \quad \text{SD}(X) \approx 1.71.$$

So a die roll typically lands about 1.7 away from its mean of 3.5.

How two variables move together

Often we care about the relationship between two quantities — study time and test score, or the firing of two neurons.

Definition

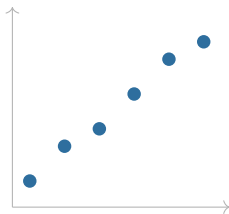
The **covariance** measures whether they tend to be large or small together:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)],$$

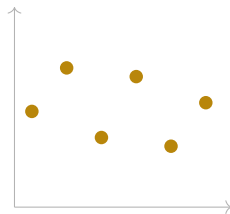
and the **correlation** rescales it to lie in $[-1, 1]$: $\rho = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}.$

Covariance is positive when the two are usually on the same side of their means, negative when one is high while the other is low.

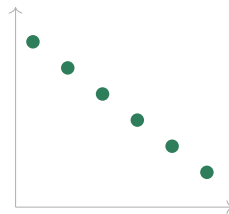
What correlation looks like



$$\rho \approx +0.9$$



$$\rho \approx 0$$



$$\rho \approx -0.9$$

Correlation measures the strength and direction of a *linear* trend: +1 a perfect rising line, 0 no linear trend, -1 a perfect falling line.

Covariance is the bridge to linear algebra

Key idea

When we have many variables at once, all their pairwise covariances are collected into a **covariance matrix**. That single object is the heart of Principal Component Analysis in Session 4. Keep the word in mind — it returns.

This is one of the places where the two halves of the course quietly become the same subject: a statistical quantity (covariance) turns into a geometric object (a matrix we can rotate and decompose).

Example

Five students' weekly study hours $x = (1, 2, 3, 4, 5)$ and quiz scores $y = (2, 4, 5, 4, 5)$.

The means are $\bar{x} = 3$, $\bar{y} = 4$. Summing the deviations gives

$$\sum (x - \bar{x})(y - \bar{y}) = 6, \quad \sum (x - \bar{x})^2 = 10, \quad \sum (y - \bar{y})^2 = 6.$$

So the correlation is

$$\rho = \frac{6}{\sqrt{10 \times 6}} = \frac{6}{\sqrt{60}} \approx 0.77 :$$

a strong positive association. (We reuse these same sums for the regression line at the end of today.)

A correlation warning

Common pitfall

$\rho = 0$ means no *linear* relationship — not “no relationship”. Points on a symmetric parabola $y = x^2$ have zero correlation despite being perfectly predictable. And correlation is never, by itself, evidence of *causation*.

In a field full of observational data, this is worth repeating often: a strong correlation is an invitation to investigate, not a conclusion about cause.

Practice — try it

A variable X takes the values $\{2, 4, 6\}$, each with probability $\frac{1}{3}$.

1. Find $\mathbb{E}[X]$.
2. Find $\text{Var}(X)$ using the computational formula.
3. If $Y = 3X$, what is $\text{SD}(Y)$?

Two theorems behind all of statistics

Why can we trust an average of noisy measurements? Two results answer this.

Key idea

Law of Large Numbers: as the sample size n grows, the sample mean $\bar{X}_n$ converges to the true mean μ . *Averages settle down.*

Central Limit Theorem: for large n , the sample mean is approximately Normal, $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$, *whatever the shape of the original data. Averages become bell-shaped.*

Why the Central Limit Theorem is remarkable

The Law of Large Numbers is intuitive: more data, better estimate. The Central Limit Theorem says something stronger and more surprising.

Even if individual observations are wildly non-Normal — a skewed reaction time, a 0/1 correct response — their *average* over many trials traces out a smooth Normal curve, and its spread shrinks in a precise way, like $\sigma/\sqrt{n}$.

This is why the Normal distribution is unavoidable, and why error bars have the shape they do.

Averaging as noise reduction

Link to cognitive science

The Central Limit Theorem underwrites the everyday practice of *trial averaging* in EEG and fMRI. Averaging n noisy trials cuts the noise standard deviation by a factor of $\sqrt{n}$, pulling a buried signal out of the noise.

It also explains why so many measured traits look Gaussian: each is the aggregate of many small independent contributions — many neurons, many trials, many genes and experiences behind a single behavioural score.

The $\sqrt{n}$ law of averaging

Useful fact

If $X_1, \dots, X_n$ are independent with mean μ and standard deviation σ , the sample mean $\bar{X}$ has

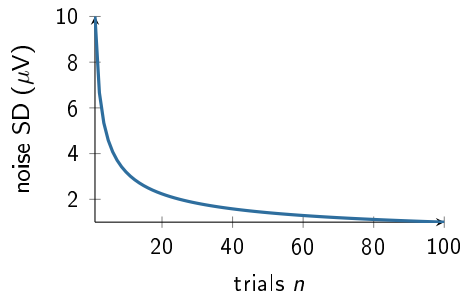
$$\mathbb{E}[\bar{X}] = \mu, \quad \text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

The key consequence is the *diminishing return*: to *halve* the uncertainty of an estimate you need *four times* as much data; to cut it tenfold, a hundredfold. This is why large studies are expensive.

The $\sqrt{n}$ law — picture and example

Example

Single-trial EEG responses have noise SD $\sigma = 10 \mu\text{V}$. Averaging 25 trials gives noise $10/\sqrt{25} = 2 \mu\text{V}$. To reach $1 \mu\text{V}$ we need $10/\sqrt{n} = 1$, i.e. $n = 100$ — four times as many trials.



Estimating what we cannot see

We almost never observe a whole population; we observe a sample and use it to guess.

Definition

A **parameter** is a fixed unknown describing the population (e.g. the true mean μ). An **estimator** is a recipe applied to the sample to approximate it (e.g. the sample mean $\bar{X}$). It is **unbiased** if it is right *on average* across many samples: $\mathbb{E}[\hat{\theta}] = \theta$.

Why we divide by $n - 1$

Common pitfall

The sample variance divides by $n - 1$, not n :

$$s^2 = \frac{1}{n - 1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Using the sample mean $\bar{x}$ (instead of the unknown true mean) makes the deviations come out slightly too small. Dividing by $n - 1$ rather than n corrects exactly for this, giving an unbiased estimate. Students should know both the rule and the reason.

Standard deviation vs. standard error

This distinction is subtle and constantly misused — so it gets its own slide.

Common pitfall

The **standard deviation** s describes the spread of the raw data; it does *not* shrink with more data. The **standard error** $SE = s/\sqrt{n}$ describes the precision of our *estimate of the mean*; it *does* shrink as n grows.

When a figure shows error bars, always ask which one they are: the spread of individuals, or the uncertainty in the mean? They tell very different stories.

Reporting uncertainty: confidence intervals

A single estimate hides its own uncertainty. A confidence interval reports a *range* of plausible values.

Definition

A 95% confidence interval for the mean is

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}.$$

The width $1.96 \sigma / \sqrt{n}$ comes straight from the Central Limit Theorem and the 95% mark of the Normal curve. The recipe is built so that, across many repeated samples, 95% of the intervals it produces contain the true μ .

When σ is unknown: use t , not z

Useful fact

In practice we estimate σ by the sample SD s . This extra uncertainty means the standardised mean follows *Student's t -distribution* with $n - 1$ degrees of freedom — like the Normal, but with heavier tails.

For small samples the 95% multiplier is larger than 1.96 (about 2.26 when $n = 10$), and it shrinks toward 1.96 as n grows. A practical rule of thumb: once $n \gtrsim 30$, the difference hardly matters.

Confidence interval — worked example

Example

A test with known population SD $\sigma = 15$ is given to $n = 25$ people; the sample mean is $\bar{x} = 105$.

The standard error is $SE = 15/\sqrt{25} = 15/5 = 3$, so the 95% interval is

$$105 \pm 1.96 \times 3 = 105 \pm 5.88, \quad \text{i.e. about } [99.1, 110.9].$$

We would report: our best estimate of the population mean is 105, and values from roughly 99 to 111 are plausible given this sample.

What “95% confidence” really means

Common pitfall

A 95% confidence interval does *not* mean “there is a 95% probability the true mean lies in *this* interval”. In the classical framework μ is fixed; it is the interval that is random.

The correct reading is about the *procedure*: if we repeated the whole experiment many times, 95% of the intervals we build this way would capture the truth. Getting this subtlety right is part of being a careful analyst.

Practice — try it

1. A sample of $n = 100$ has mean $\bar{x} = 52$ with known $\sigma = 10$. Give a 95% confidence interval for μ .
2. Testing $H_0 : \mu = 50$ against $\mu \neq 50$, what is the z-statistic, and the rough conclusion?
3. A treatment raises scores by 2 points, with within-group SD 10. What is Cohen's d , and is it small, medium, or large?

The logic of a hypothesis test

The dominant — and most criticised — ritual of statistics. Students must be able to both perform it and understand its limits.

Four steps

- (1) State a **null hypothesis** H_0 (“nothing is going on”) and an alternative H_1 .
- (2) Choose a **test statistic** that would be extreme if H_1 were true.
- (3) Compute the **p -value**: assuming H_0 , the probability of a result at least this extreme.
- (4) If $p < \alpha$ (often 0.05), **reject** H_0 .

Example

A coin lands heads 60 times in 100 flips. Is it fair?

Under $H_0 : p = 0.5$, the number of heads has mean $np = 50$ and standard deviation $\sqrt{np(1-p)} = \sqrt{25} = 5$. The standardised statistic is

$$z = \frac{60 - 50}{5} = 2.0.$$

A result this extreme (either tail) has probability $\mathbb{P}(|Z| \geq 2.0) \approx 0.046$. Since $0.046 < 0.05$, we reject H_0 : weak but statistically significant evidence of bias.

Definition

A **Type I error** is rejecting a true H_0 (a false positive), with probability α . A **Type II error** is failing to reject a false H_0 (a false negative), with probability β . The **power** of a test is $1 - \beta$: its chance of detecting a real effect.

There is always a trade-off: making α very small (few false positives) tends to lower power (more missed real effects). Choosing that balance is a design decision, not a mathematical one.

The workhorse tests

All follow the same four-step logic; they differ only in the statistic used.

- **t -test** — compares means (one sample, or two groups).
- **χ^2 test** — compares observed vs. expected counts in categories.
- **ANOVA** (F -test) — compares means across three or more groups.

Each carries assumptions worth stating out loud: roughly, independence of observations and approximate Normality of the relevant averages.

Significant is not the same as big

A real effect the test misses

Two groups of 25 score means 105 vs. 100, pooled SD 10. Then $SE = 10\sqrt{2/25} \approx 2.83$, $t \approx 1.77$, giving $p \approx 0.08$ — *not* significant. Yet the effect size is $d = \frac{5}{10} = 0.5$, a solidly “medium” effect. The study is simply underpowered.

The lesson: always report an *effect size* (like Cohen's d or ρ) alongside the p -value.

The p -value and the replication crisis

Link to cognitive science

The p -value is routinely misunderstood. It is *not* the probability that H_0 is true, nor that the result was “due to chance”. Misuse of $p < 0.05$ — through multiple comparisons, optional stopping, and publication bias — is a central driver of the **replication crisis** that has reshaped psychology and cognitive neuroscience since the 2010s.

Reporting effect sizes and confidence intervals, and pre-registering analyses, are part of the modern remedy.

From correlation to prediction: regression

Correlation measures the strength of a linear relationship. Regression draws the actual line and lets us *predict*.

Definition

Simple linear regression fits $\hat{y} = a + bx$ by **least squares** — choosing a, b to minimise the total squared vertical error $\sum (y_i - \hat{y}_i)^2$. The solution is

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}, \quad a = \bar{y} - b\bar{x}.$$

What least squares is doing

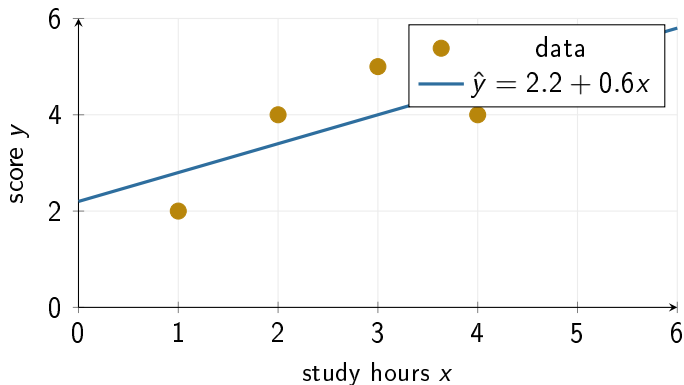
Intuition

We look for the straight line that passes as close as possible to all the points, where “close” is measured by the squared vertical gaps between each point and the line. The slope turns out to be just the covariance of x and y , rescaled by the spread of x — the same covariance from an hour ago.

Two facts that make good checks

The residuals always sum to zero and are uncorrelated with x ; and the line always passes through the mean point $(\bar{x}, \bar{y})$.

The regression line, drawn



The fitted line through the study-hours data. Each extra hour of study predicts about 0.6 more points.

Example

Using the study-hours data, with $\sum(x - \bar{x})(y - \bar{y}) = 6$ and $\sum(x - \bar{x})^2 = 10$.

$$b = \frac{6}{10} = 0.6, \quad a = 4 - 0.6 \times 3 = 2.2, \quad \hat{y} = 2.2 + 0.6x.$$

The fraction of variance the line explains is $R^2 = \rho^2 \approx 0.77^2 \approx 0.60$: the line accounts for about 60% of the variation in scores.

Regression is the atom of modeling

Link to cognitive science

Extend regression to many predictors and it becomes *multiple regression*; add a nonlinear “link” and it becomes the *Generalized Linear Model* behind psychometric functions and neural encoding models (Session 5); view it as “fit parameters to minimise error” and it is exactly the objective of supervised machine learning.

And geometrically, least squares is a *projection* — which is where linear algebra picks up the story next session.

Practice — try it

Four points: $x = (0, 1, 2, 3)$, $y = (1, 1, 2, 4)$.

1. Find $\bar{x}$ and $\bar{y}$.
2. Compute $\sum(x - \bar{x})(y - \bar{y})$ and $\sum(x - \bar{x})^2$.
3. Give the regression line $\hat{y} = a + bx$.

Session 2 — what we built

- Summaries: the *mean* (centre), *variance/SD* (spread), and *covariance/correlation* (co-movement).
- The *LLN* and *CLT*: averages converge and become Normal with spread $\sigma/\sqrt{n}$ — the basis of standard errors and confidence intervals.
- *Hypothesis testing*: *p*-values, the errors, and why significance is not importance.
- *Regression*: a least-squares line, reusing covariance and pointing ahead to the GLM and machine learning.

Next: linear algebra — vectors, matrices, and geometry.